

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-SMS-015

Title: Hot Metal Ladle Handling and Transfer

Department: Steel Melting Shop

Revision: 01

Date: 18-Mar-2024

1. PURPOSE

To ensure safe handling, lifting and transfer of hot metal ladles between the mixer, desulphurisation unit and LD converters.

2. SCOPE

Applies to all ladle cranes, crane operators, riggers and floor personnel in the charging bay.

3. LADLE CONDITION CHECKS (BEFORE EVERY USE)

- Refractory lining visually inspected; no red spots or bulging on the shell
- Trunnions and lifting hooks checked for cracks; last NDT not older than 6 months
- Ladle must be pre-heated to minimum 900°C if it has been idle for more than 2 hours
- Ladle must be completely dry; a wet or damp ladle must NEVER be filled

4. LIFTING AND TRANSFER

- Only a certified ladle crane (125 t capacity) may be used
- Maximum fill level: 300 mm below the ladle rim
- No person shall stand or pass under a suspended ladle
- Maintain an exclusion zone of 10 metres around the ladle path during transfer
- Travel speed must not exceed 30 m/min while carrying a full ladle
- Signal person with whistle and flag must guide every move

5. PPE

- Aluminised apron, heat-resistant gloves and face shield for anyone within 10 metres
- Safety boots with metatarsal guard

6. EMERGENCY — LADLE BREAKTHROUGH OR SPILL

Step 1 — Sound the bay siren and move everyone out of the bay immediately.

Step 2 — Do NOT use water on molten metal.

Step 3 — Crane operator to place the ladle in the nearest emergency pit if safe to do so.

Step 4 — Call Emergency Control Room: Ext. 100 and Fire Station: Ext. 101.

Step 5 — Allow spilled metal to solidify before any clean-up; use dry sand only.